



ScopeX Advanced Oscilloscope

Setup Manual

Driver Installation & Host Software Framework

ScopeX Integrated Laboratory Framework

- Dual-Channel Digital Storage Oscilloscope (DSO)
- Integrated DDS Analog Signal & Digital Frequency Generator
- Fixed +5VDC and High-Precision Bipolar $\pm 12V$ Dual Rail Supply
- Cross-Platform Native GUI Deployment (Windows, macOS, Linux)
- Multi-OS USB Communications Class Device Architecture (CDC-ACM)

Revision Tracking: v3.0

Supported Architecture Nodes: Windows 10/11 · macOS Sierra+ · Linux (Ubuntu 20.04+)

Confidential — For internal, industrial, and educational use

TABLE OF CONTENTS

Contents

1	Introduction	2			
1.1	Cross-Platform Core Infrastructure	2			
1.2	Kit Components Inventory	2			
2	System Setup and Configuration	3			
2.1	Initial Physical Layout Logic	3			
2.2	Windows Setup Procedure	3			
2.2.1	Install Application and Drivers	4			
2.2.2	Open Executable	5			
2.2.3	Connect Scopex and Find Device	6			
2.2.4	Update Driver	7			
2.2.5	Install Driver	8			
2.2.6	Verify COM Port Recognition	9			
2.2.7	Launch GUI and Check Ports	10			
2.3	macOS Setup Procedure	11			
2.3.1	Mount and Unpack Application Bundle	12			
2.3.2	Bypass Gatekeeper Execution Restrictions	14			
2.3.3	Initialize Host Interface Node Alignment	16			
2.4	Linux Setup Procedure	17			
2.4.1	Configure Dialout Communication Privileges	18			
2.4.2	Assign Program Execution Flags	20			
2.4.3	Confirm ACM Endpoint Connection Mapping	22			
3	Hardware Architecture	23			
3.1	Main Board Overview	23			
3.2	Board Features	24			

1 Introduction

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Overview: The Scope-X reference hardware system from Bumblebee Instruments relies on an onboard ATxmega processor executing a standard USB Communications Device Class (CDC-ACM) profile. This protocol ensures reliable Virtual COM Port mapping across high-speed structural links.

This manual details the procedures required to flash, map, and bind host drivers to the Scope-X GUI interface application framework.

1.1 Cross-Platform Core Infrastructure

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Depending on your structural target host platform (Windows, macOS, or Linux), device node acquisition requires specific local terminal access privileges. Users are advised to retrieve the structural workspace bundle containing the host executable binaries and the verbatim archive file named `Drivers.zip` before proceeding with initialization strings.

- **Windows 10/11 Deployment:** Utilizes inf-driven virtual structural mapping to allocate system terminal objects.
- **macOS Architectures:** Operates via the built-in communication class system layer, utilizing Gatekeeper pathway authorization.
- **Linux Environments:** Relies on kernel-level `cdc_acm` modules requiring structural group permission modifications.

1.2 Kit Components Inventory

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Table 1: System Hardware Packaging Profile

Component Description			Interface Specification	Quantity
Scope-X Board	Main Processing		ATxmega Core / Analog Signal Mesh Layer	1
Power Management Link	Management Interface		Regulated Rail Interface Cable	1
High-Speed Comm-Link	Comm-Link	Cable	Custom Shielded USB-B Link Cable	1

2 System Setup and Configuration

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Important Deployment Rules: Observe clean Electrostatic Discharge (ESD) procedures when manipulating the processing architecture. Ensure the host processing terminal environment satisfies the cross-platform dependencies listed below.

2.1 Initial Physical Layout Logic

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1. Unpack the structural hardware component stack and evaluate it against the packaging layout documentation.
2. Extract the software workspace folder onto a verified local storage matrix tracking route.
3. Provide the hardware block with proper power via the tracking cable. Verify active LED status loops.
4. Secure the USB-B link to target structural host ports to register internal endpoints.

Host Terminal System Requirements

- **Windows Platform:** Windows 10 / 11 (x64 Architecture)
- **Apple Platform:** Intel / Apple Silicon (M1/M2/M3) macOS 10.12+
- **Linux Open-Source:** Kernel 5.4+ (Ubuntu 20.04 LTS or equivalent)
- **Physical Port Allocation:** Dedicated USB 2.0/3.0 Type-A/Type-C Slot

2.2 Windows Setup Procedure

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2.2.1 Install Application and Drivers

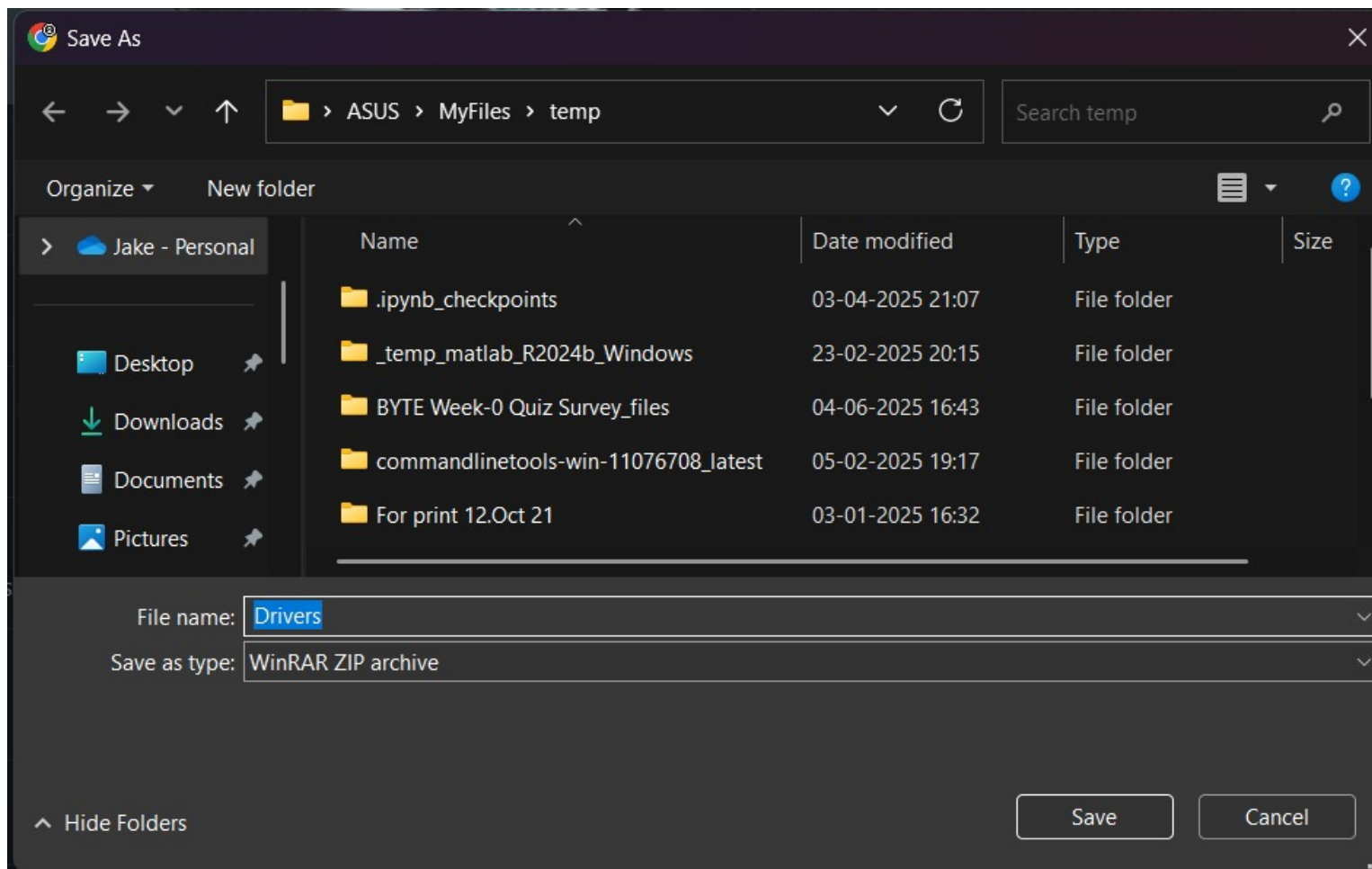


Figure 1: Locate your workspace bundle. Save the verbatim **Drivers.zip** archive and extract its contents to your preferred system directory alongside the main executable binary.

2.2.2 Open Executable



Figure 2: Open the host executable application. Select 'Run anyway' if prompted by the Windows SmartScreen security filter.

2.2.3 Connect Scopex and Find Device

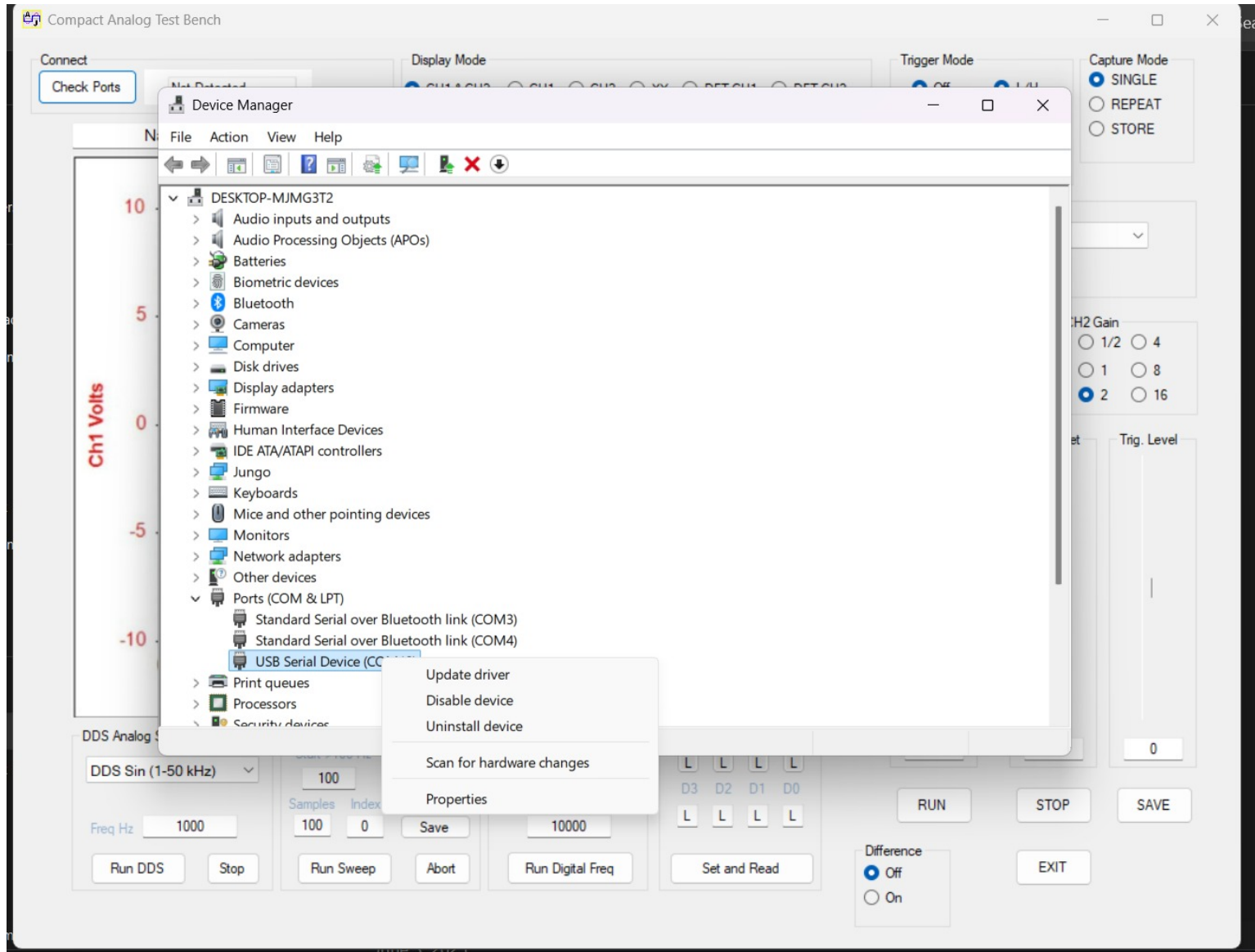


Figure 3: Interface the core board via USB. Open Device Manager and locate the unmapped 'USB Serial Device' entry inside the Ports (COM & LPT) categorization path.

2.2.4 Update Driver

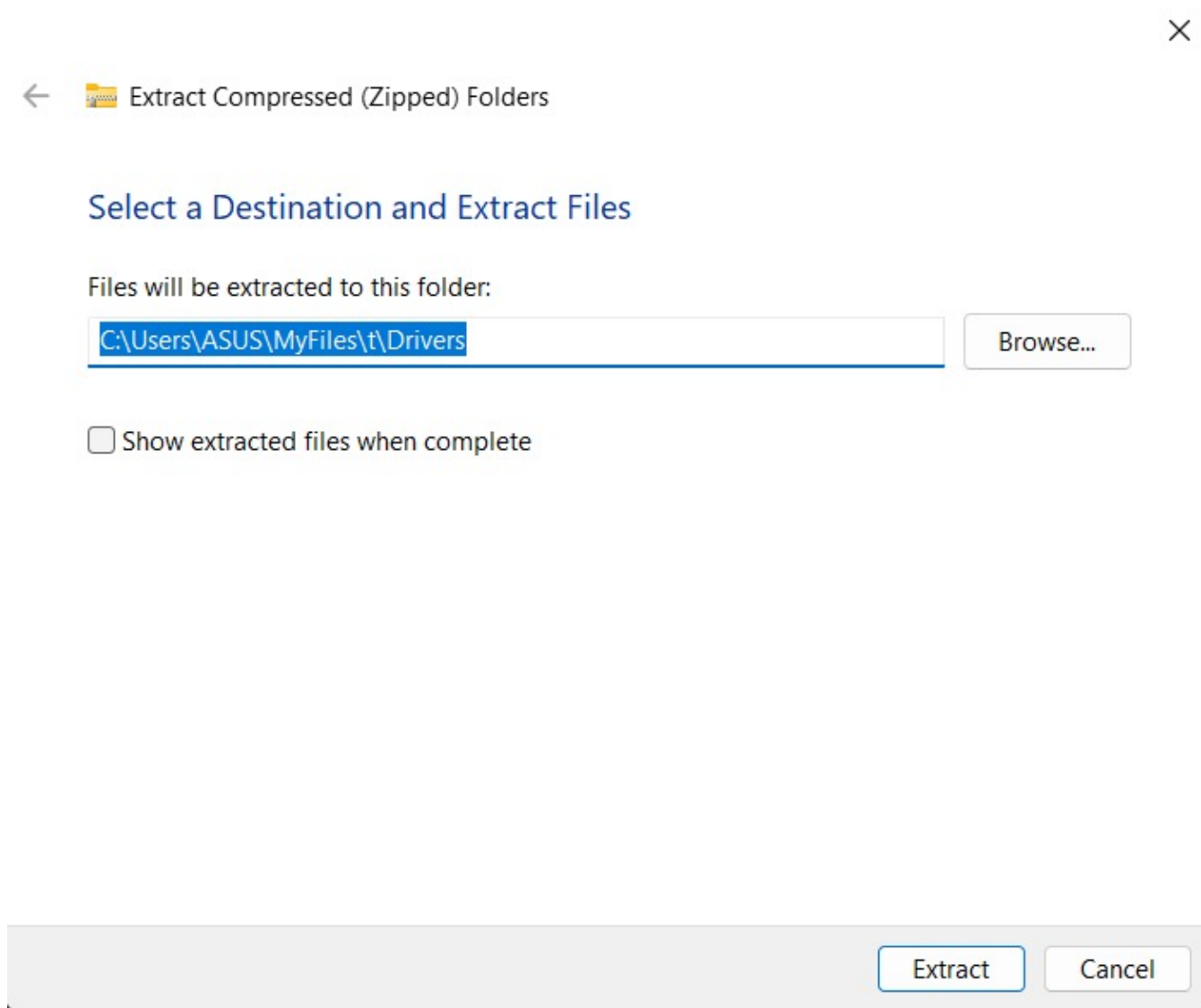


Figure 4: Right-click on the device entry, choose 'Update Driver', select the manual local lookup parameter, and navigate to the directory containing the uncompressed structure from `Drivers.zip`.

2.2.5 Install Driver

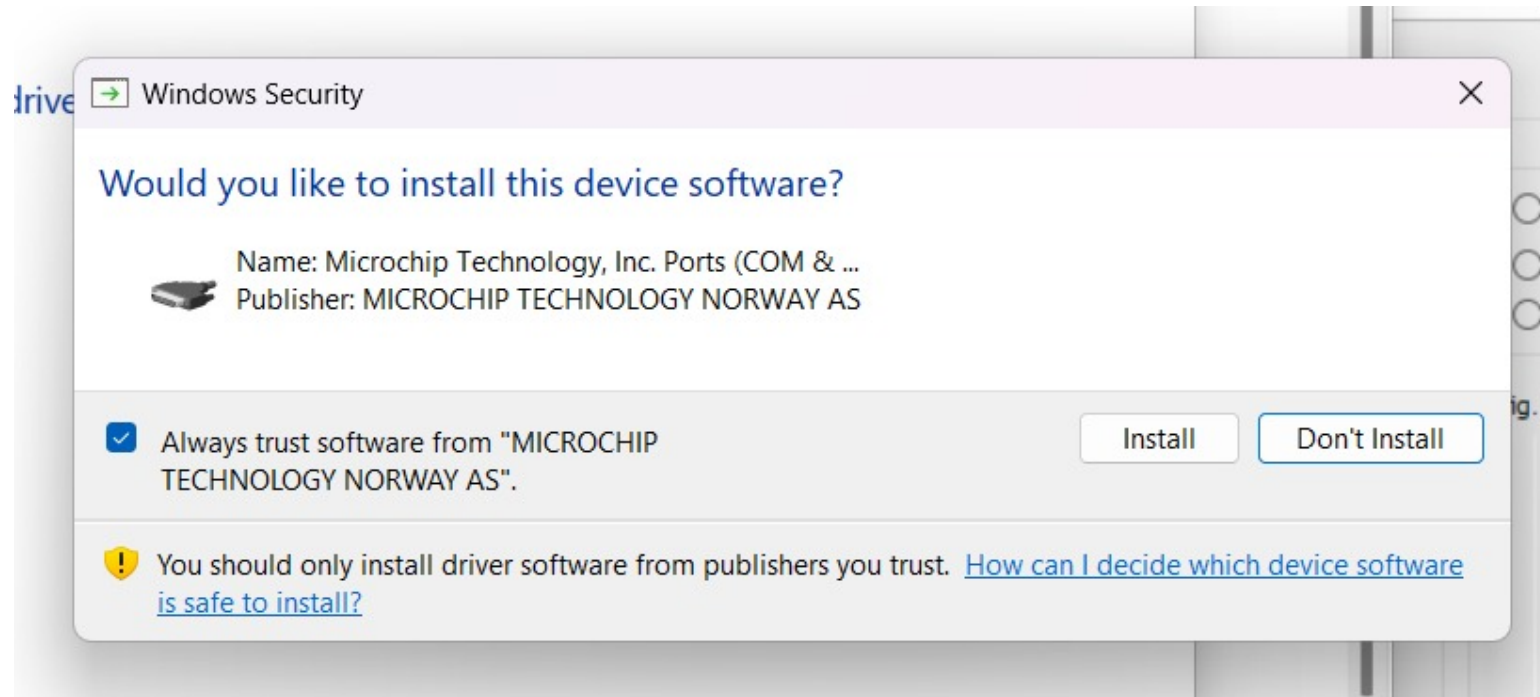


Figure 5: Accept any system prompts to authorize initialization. The tracking engine registers the inf parameters into host infrastructure indexes.

2.2.6 Verify COM Port Recognition

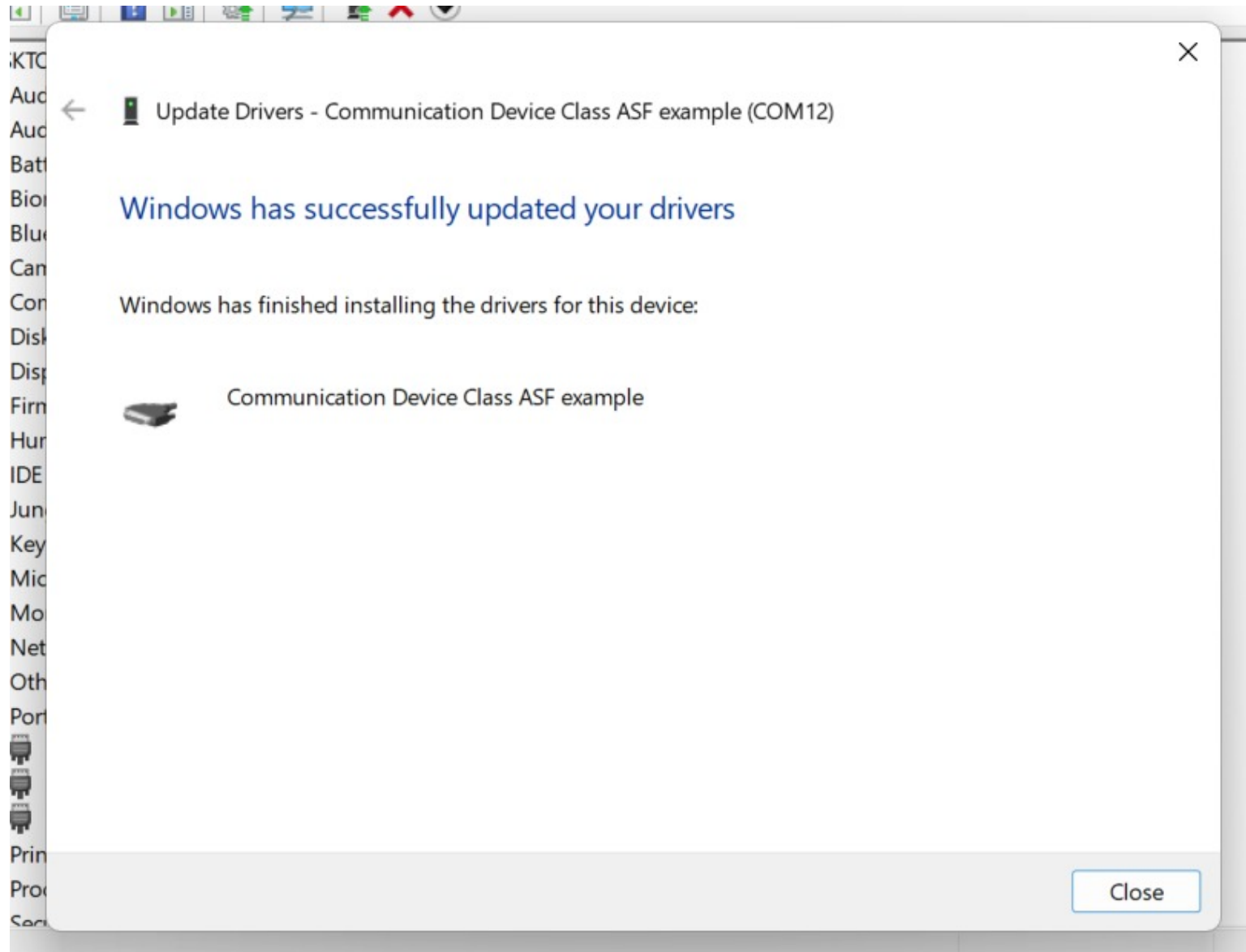


Figure 6: Verify that the system architecture successfully maps the target hardware, re-indexing it as a verified virtual communication endpoint node under an assigned COM identification block.

2.2.7 Launch GUI and Check Ports

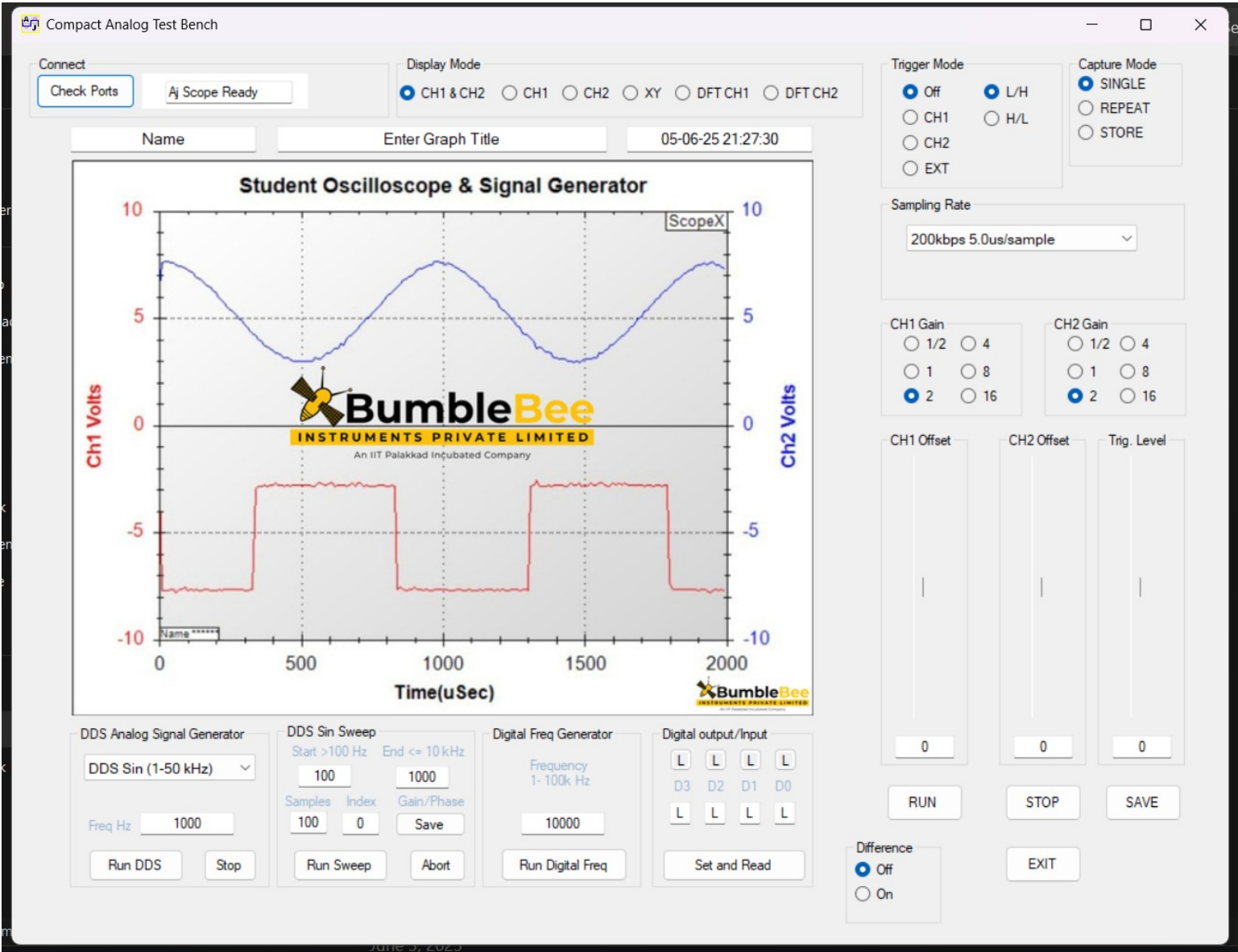


Figure 7: Launch the graphical layout tool, process a 'Check Ports' request token, and confirm the active device handshake state.

2.3 macOS Setup Procedure

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Modern macOS platforms automatically intercept ATxmega USB targets using the internal communications layer (**AppleUSBACM**). Driver setup bypasses manual file configuration and instead focuses on passing host execution policies.

1. Extract the application bundle and files on your Mac system.
2. Direct the uncompressed workspace structure into a safe tracking

folder destination.

3. Attach the Scope-X board to provide input to the system serial descriptor arrays.
4. Override system pathways blockers to ensure structural processing stability.

2.3.1 Mount and Unpack Application Bundle

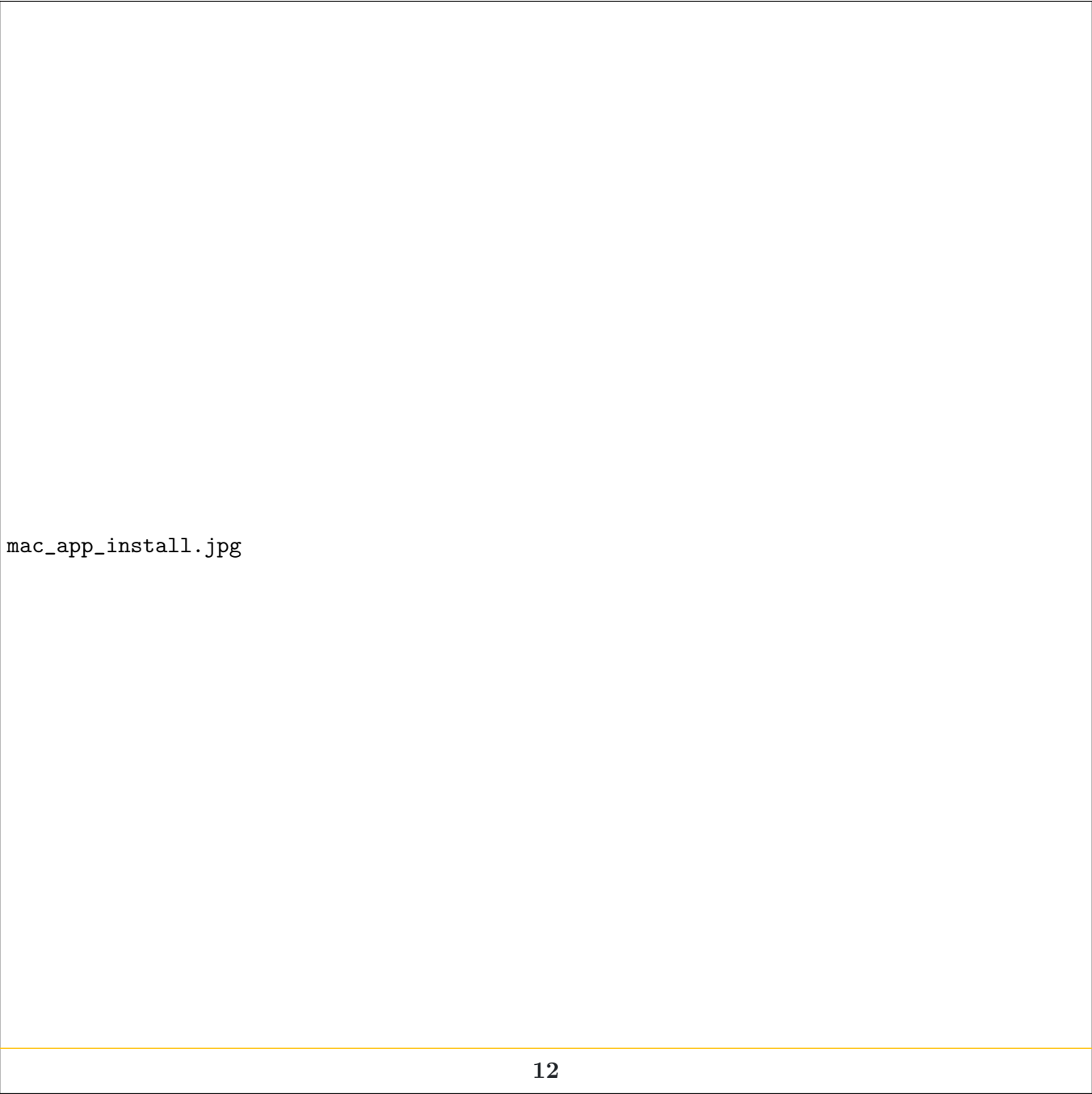
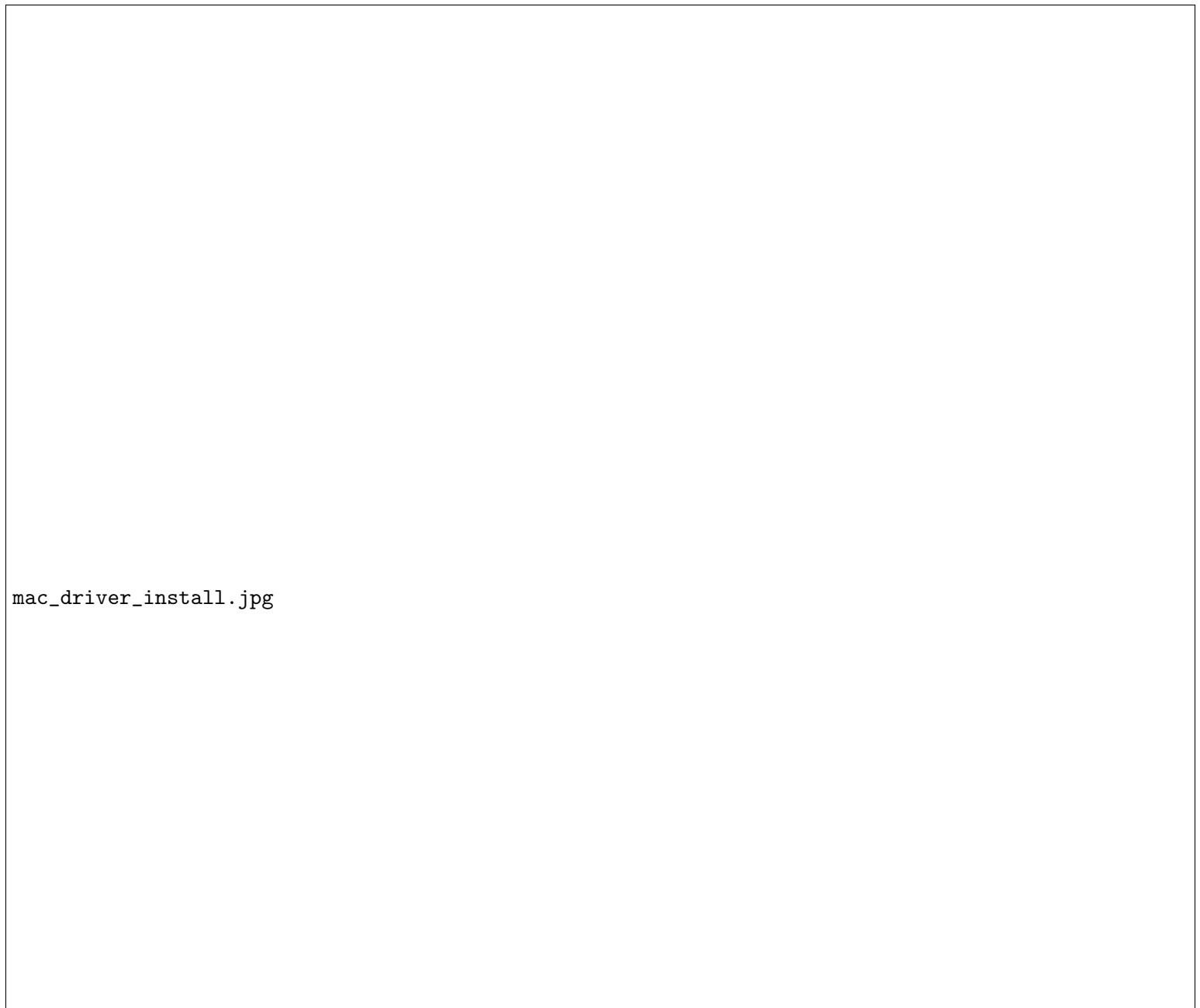


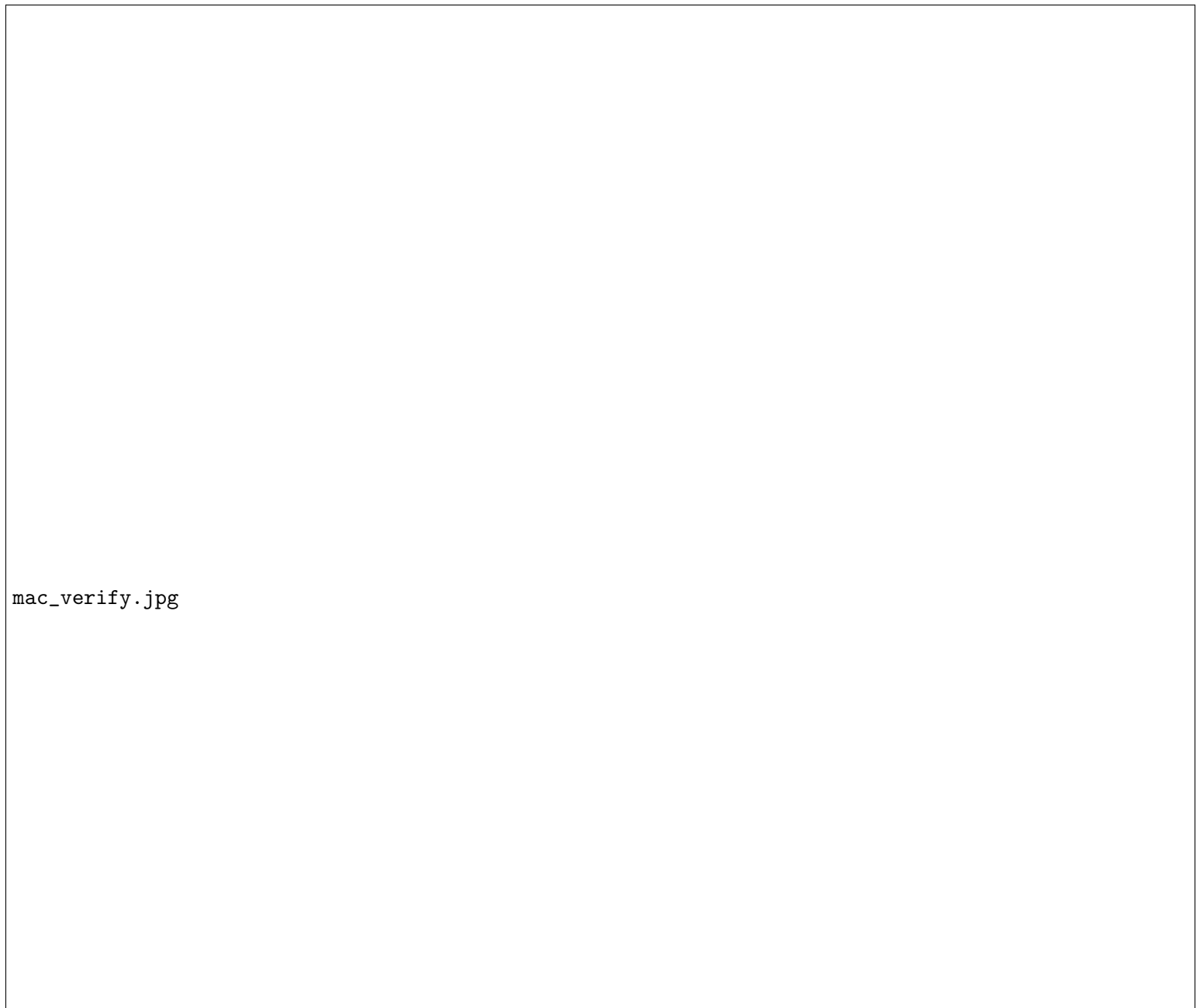
Figure 8: Unpack the Scope-X application structure and store the uncompressed tracking binary components into your local operational

2.3.2 Bypass Gatekeeper Execution Restrictions



mac_driver_install.jpg

2.3.3 Initialize Host Interface Node Alignment



mac_verify.jpg

2.4 Linux Setup Procedure

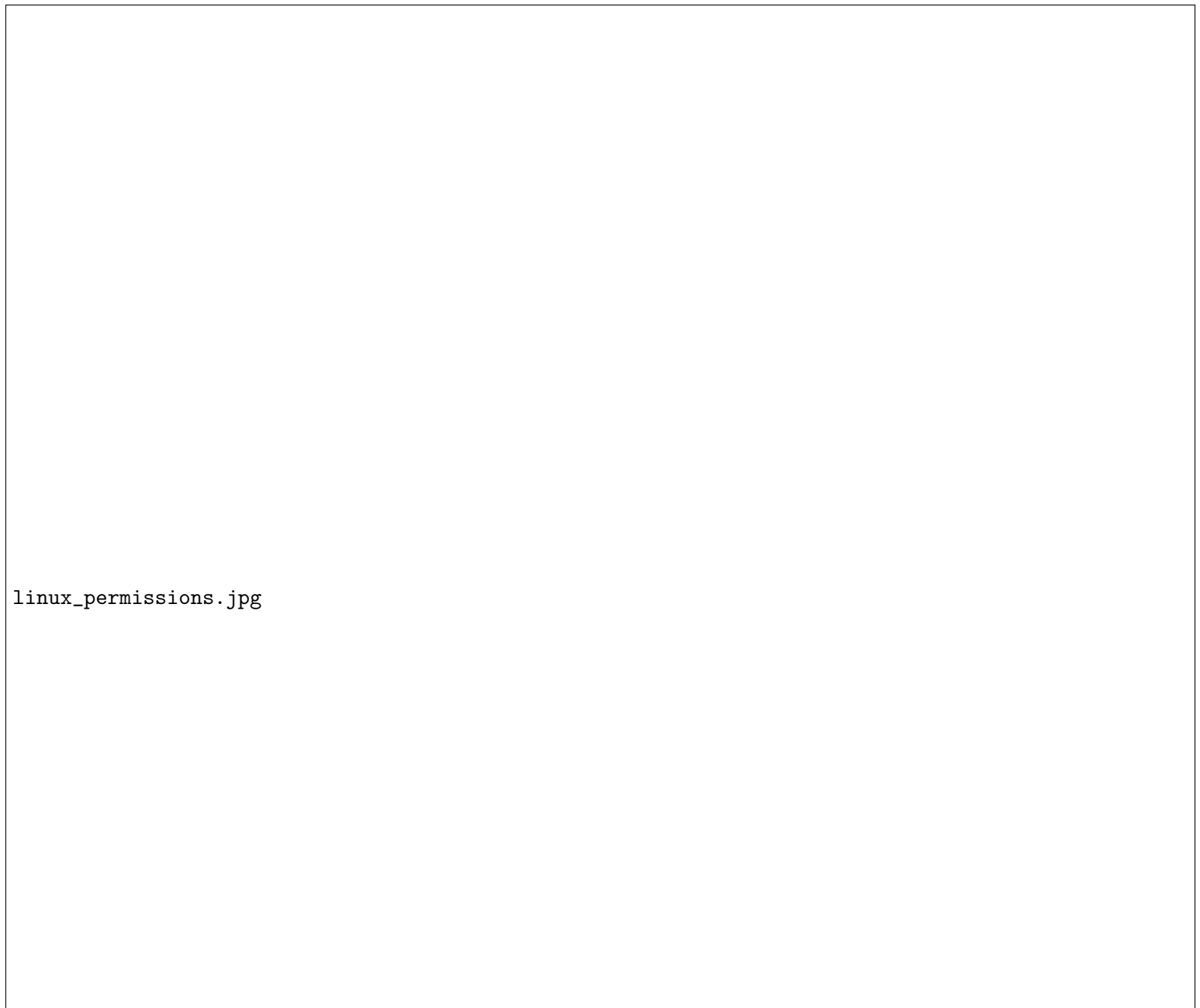
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Linux systems use the kernel-level `cdc_acm` module to capture ATxmega hardware links. While port mapping occurs immediately upon insertion, character node entry boundaries (`/dev/ttyACM*`) require manual modification of group permissions.

1. Extract the target cross-platform application structure folder into your project directory workspace.
2. Map localized user system descriptors into the operational serial interface security profiles.
3. Re-index file mode flags to enable program deployment behavior for the binary core.
4. Reset user permissions inside the shell environment and initialize system monitoring loops.

2.4.1 Configure Dialout Communication Privileges



linux_permissions.jpg

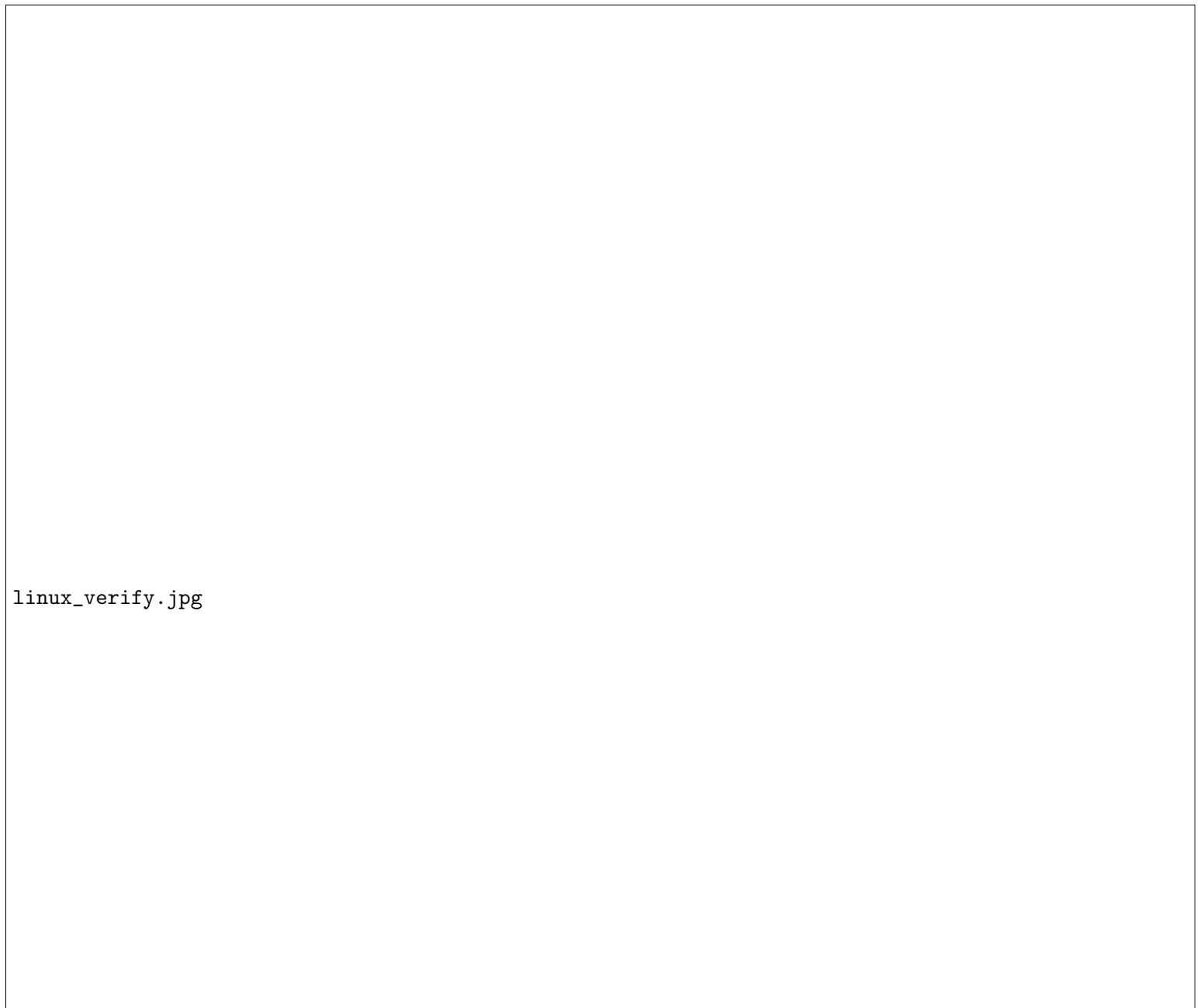
2.4.2 Assign Program Execution Flags



linux_run.jpg

Figure 12: Access file parameters locally or enter this command into your terminal: `chmod +x ScopeX_GUI` to grant execution permission to

2.4.3 Confirm ACM Endpoint Connection Mapping



linux_verify.jpg

3 Hardware Architecture

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3.1 Main Board Overview

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The Scope-X main board features a modular architecture optimized for educational use and rapid prototyping.

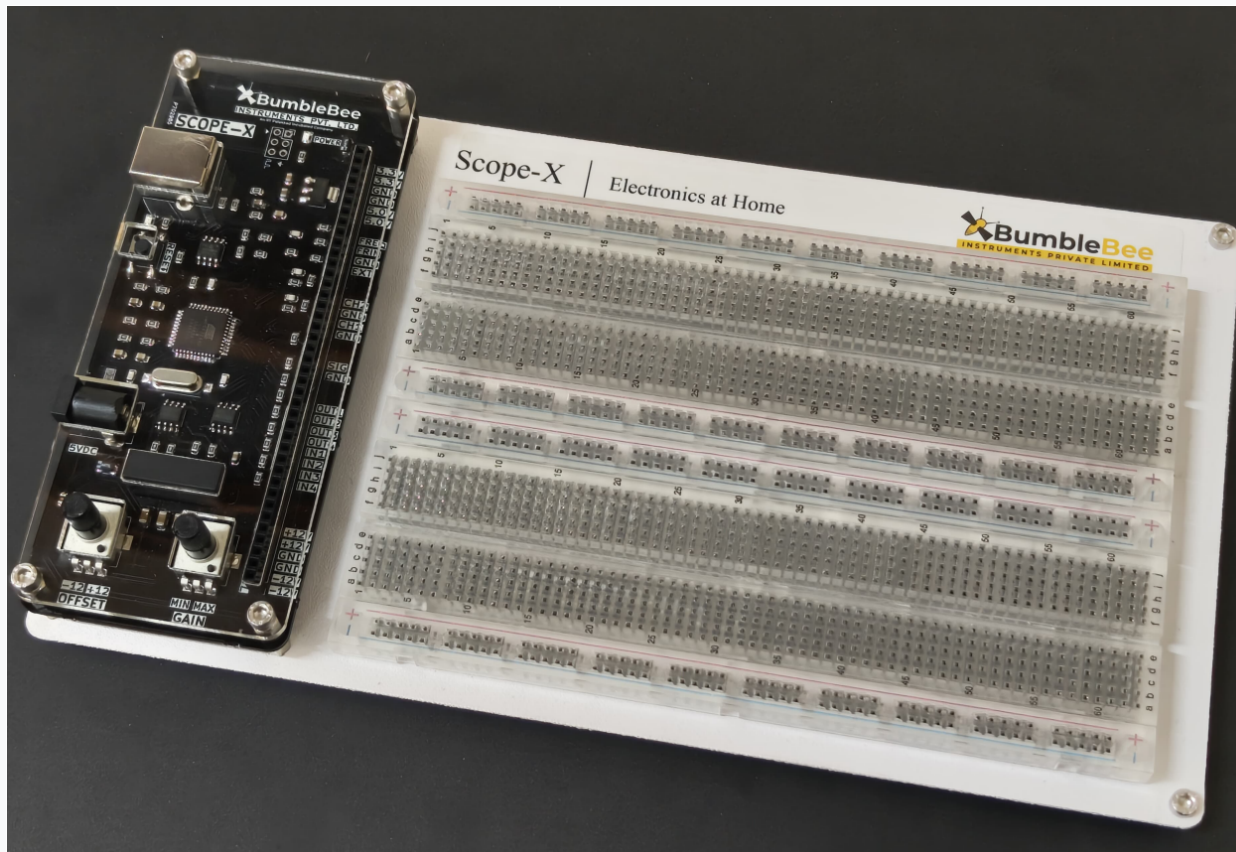


Figure 14: Scope-X Main Board Layout Schema

3.2 Board Features

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Figure 15: Scope-X Processing Hardware Subsystem

Table 2: System Mapping Allocation Table

Feature	Assignment	Operational Use Case Parameters
1	USB-B Port	High-Speed Virtual COM Pipeline
2	Power Switch	Main Rail Isolation Management
3	Reset Link	Core Microcontroller Restart Matrix
4	External Bus	Accessory Processing Routing Matrix
5	Scope Inputs	Dual-Channel System Diagnostics Block
6	DDS Generator	Programmable Continuous Waveform Matrix
7	5VDC Rail	System Logic Board Supply Link
8	Digital I/O Block	4-Bit Capture and Stimulation Channels
9	Offset Adjust	Operational Tuning Block (-12V to +12V)
10	Gain Matrix	Precision Hardware Scaler Interface
11	Dual Rail Block	Bipolar Lab Reference Subsystem